DIAN RAISA GUMILAR

Informatics Engineering | Artificial Intelligence & Machine Learning | Data Science & NLP

Neglasari, Cibeunying Kaler, Kota Bandung, Jawa Barat • dian.gumilar04@gmail.com • +62 8122 1409 559 •

linkedin.com/in/dian-raisa-gumilar • dianraisagumilar.vercel.app

PROFESSIONAL SUMMARY

Informatics graduate from Institut Teknologi Nasional (Itenas) Bandung specializing in Artificial Intelligence and Machine Learning. Proficient in Python-based data processing, Natural Language Processing (NLP), and predictive model evaluation. Demonstrates strong analytical thinking, systematic problem-solving, and cross-functional collaboration, eager to drive impactful software and AI solutions.

PENDIDIKAN

Institut Teknologi Nasional Bandung — S1 Teknik Informatika (IPK: 3.58 / 4.00) **2022 – 2026**

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Bachelor of Informatics graduate from Institut Teknologi Nasional (Itenas) Bandung with a 3.58 GPA (Cum Laude), focusing on Artificial Intelligence, Machine Learning, and Deep Learning. Proven academic contribution as a Machine Learning Laboratory Assistant and Computer Networks Teaching Assistant, guiding students through practical exercises and technical problem-solving. Actively engaged in various IT and technology competitions, notably achieving 1st Place in the Huawei ICT Competition National Final (Computing Track). Holds an internationally recognized Huawei Certified ICT Associate (HCIA-AI V3.5) certification with a score of 960/1000. Research capabilities are demonstrated by two international publications indexed in IEEE and Emerald Publishing focusing on BERTopic, as well as an officially registered Software Copyright (HKI) from DJKI.

- **Himpunan Mahasiswa Informatika (HMIF) Itenas Bandung - Staff of Internal Audit Unit (Jul 2024 – Dec 2024):** Monitored organizational governance, operational procedures, and program compliance across student association divisions to ensure operational transparency and accountability. Conducted performance evaluations and administrative audits on student work programs, delivering formal evaluation reports to leadership boards.
- **Himpunan Mahasiswa Informatika (HMIF) Itenas Bandung - Staff of Cadre Division / Divisi Kaderisasi (Sep 2023 – Jun 2024):** Co-designed and executed character-building curricula and orientation programs aimed at accelerating cultural adaptation for incoming informatics freshmen. Facilitated leadership development workshops, collaborative activities, and mentorship initiatives to nurture team synergy and active student involvement.

PENGALAMAN KERJA

Asisten Laboratorium Machine Learning — Institut Teknologi Nasional Bandung **Februari 2026 – Juni 2026**

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- Facilitated hands-on laboratory sessions for undergraduate students across 10 structured modules, guiding end-to-end data pipelines including data preprocessing (handling missing values, outliers, encoding, and normalization), exploratory analysis, and model evaluation.
- Leveraged Python, Pandas, Matplotlib, and Scikit-Learn to deliver live technical demonstrations on dimensionality reduction, clustering metrics, and machine learning algorithms such as Naïve Bayes, regression, and comparative classification models.
- Resolved technical coding and debugging issues for 50+ cohort students, while providing direct hands-on mentorship to a group of 8 students, successfully guiding them to top academic results.

Asisten Dosen Jaringan Komputer — Institut Teknologi Nasional Bandung **April 2026 – Mei 2026**

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- Delivered technical instruction and guided hands-on lab sessions as a substitute teaching assistant, stepping in to ensure uninterrupted learning continuity for the Computer Networks course.
- Taught core network layer implementations, including Virtual LAN (VLAN) segmentation, inter-VLAN routing, and Open Shortest Path First (OSPF) dynamic routing protocol configurations.
- Utilized network simulation tools (eNSP) and CLI environments to facilitate live configuration demonstrations, assist students with practical debugging, and resolve setup errors.

Penelitian & Pengembangan / Magang IT — BRIN - Badan Riset dan Inovasi Nasional **Maret 2024 – Juni 2024**

Dago, Coblong, Kota Bandung, Jawa Barat, Indonesia

- Conducted Python-based text processing and topic modeling pipelines using BERTopic, NLP techniques, and data manipulation libraries (Pandas, NumPy) to analyze academic publication datasets for researcher-group recommendations.
- Developed, tested, and validated an analytical research prototype ("Collab Identifier") to discover cross-institutional thematic research collaboration patterns.
- Delivered high-impact research outcomes, resulting in an officially registered software copyright (HKI) from DJKI and an international journal publication in Emerald Publishing (Library Hi Tech).

PELATIHAN

Huawei Course Basics of Network Layer Protocols — HUAWEI (Teknologi Informasi dan Komunikasi) **Oktober 2025 – Desember 2025**

Key Learnings: Fundamental network infrastructure concepts, IPv4/IPv6 addressing schemes, packet forwarding mechanisms, and routing protocol operation.

Impact & Outcome: Built a robust technical foundation for diagnosing packet-level network bottlenecks, troubleshooting routing anomalies, and managing complex multi-subnet network environments.

Huawei ICT Associate (HCIA) AI V3.5 — Huawei ICT Associate (Teknologi Informasi dan Komunikasi) **Juli 2024 – Agustus 2024**

Key Learnings: Machine Learning and Deep Learning architectures, Python predictive modeling, MindSpore framework, ResNet-50 image classification, and end-to-end cloud training pipelines using ModelArts.

Impact & Outcome: Achieved an outstanding score of 960/1000, demonstrating certified readiness to develop, evaluate, and deploy scalable computer vision and AI solutions in enterprise production environments.

PENCAPAIAN

HKI (Hak Kekayaan Intelektual) Program Komputer - Collab Identifier — Direktorat Jenderal Kekayaan Intelektual (DJKI) **Juni 2025**

Awarded official intellectual property protection for the computer software "Collab Identifier", an analytical platform bridging NLP and Library & Information Science (LIS) to evaluate cross-institutional research synergies.

Juara 1 Huawei ICT Competition 2025-2026 Tingkat Nasional — Huawei **Desember 2025**

Outperformed national university teams across Indonesia by solving advanced networking and cloud-computing challenges, demonstrating strong technical execution and problem-solving under competition environments.

Publikasi Ilmiah Internasional Emerald: Collab Identifier using BERTopic — Emerald Publishing (Library Hi Tech) **Oktober 2026**

Authored the peer-reviewed research paper "Collab Identifier: A Prototype for Uncovering Thematic Research Collaboration Opportunities" in an internationally indexed journal, introducing an NLP- and BERTopic-driven framework to discover inter-institutional research partnerships.

Publikasi Ilmiah Internasional IEEE: Integrating BERTopic and Heterogeneous Knowledge Graphs — IEEE (The 13th IC3INA) **Oktober 2026**

Co-authored and presented "Integrating BERTopic and Heterogeneous Knowledge Graphs for Temporally-Evaluated Research Collaboration Recommendation" at the IEEE IC3INA conference, designing a temporal link prediction architecture for scientific publications.

Graduate with Honors (Cum Laude) – Bachelor's Degree in Informatics — Institut Teknologi Nasional Bandung **Oktober 2026**

Completed undergraduate studies on time with an honors grade point average of 3.58/4.00, recognized for solid academic standing and research contributions in artificial intelligence.

SERTIFIKASI

Huawei ICT Associate (HCIA) AI V3.5 — **Huawei** (Skor: 960) **Agustus 2024 – Agustus 2027**

Huawei Course Basics of Network Layer Protocols — **Huawei** (Skor: 0) **Desember 2025 – Desember 2027**

KEAHLIAN

Collaboration, Problem Solving, Knowledge Graphs, Flutter, Mobile Development, Computer Networks, IoT, Teamwork, Multimedia, Machine Learning, Artificial Intelligence (AI), PyTorch, Time Management, Topic Modeling, Model training & evaluation, scikit-learn, Mentoring, Natural Language Processing (NLP), Computer Vision, Deep Learning, analytical thinking, Audit, Dart, SQL, Tensorflow, Networking, Python, Desktop Application Development, Databases.

BAHASA

Bahasa Indonesia: Tingkat Fasih atau Bahasa Ahli | **English (Bahasa Inggris):** Tingkat Profesional